The DB Move, Measured End to End

Geometry, Descent, Window Length, Curation — and the GSS Profile

local_mixing working notes

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Abstract

This note reports a campaign of roughly seventy fmix runs that measured the frozen-store DB move along four axes — window *geometry* (convex vs contiguous), *prefix descent*, *window length* s_{db}, and *curation* — in both COMP and MIX modes, on material ranging from fresh gadget output to near-minimal circuits. Four results dominate. (1) Convex beats contiguous on every axis in COMP: 16× the gates removed, 7× the ancestry transport, in 1/31 the wall time; the shipped default was inverted and has been fixed. (2) Prefix descent is not a size lever but a *transport* lever, worth 600× on ancestry while being 6× *worse* on size per CPU-second. (3) The MIX-DB move is a mean-reversion operator on window length with a fixed point near 5: short windows expand, long windows compress, and s_{db} is therefore a net-growth-rate knob rather than a search-width knob. (4) The curated store is a *wide-window* store — worthless at length 1 and worth 19× at length 20. We close with the GSS profile, the settings for running fmix on a gadgetized sliced sandwich, and with a section on the measurement traps this campaign walked into, which we think is the most reusable part.

Contents

1 What was measured, and on what

All runs use the frozen store `frozen_m1_m11` with the curated store `frozen_curated_m1_m11` under the `legacy-swapped-controls` value convention, `-db-advance` on throughout, span cap 30, degree cap 9.

tag	description	gates	character
`pre2_100k`	fresh gadget output	100,000	52% g57 fossils
`nR20_mixed`	profile-run output, near-minimal	179,132	99.3% g57, $d_{\min} \approx 0.002$
`cs_p20_mixed`	p_mix 0.20 growth run, never compressed	202,674	MIX-grown fat
`cs_p35_mixed`	p_mix 0.35 growth run, never compressed	317,813	more fat

Table 1: Input circuits. The last two were produced by a parallel session's p_mix sweep and are used as "lots of easily compressible fat".

2 Part I: the 32-arm COMP factorial

2.1 Design

A full 2^5 factorial on `pre2_100k`: $s_{db} \in \{12, 13\}$ × descent on/off × convex/contiguous × reshuffle on/off × curated on/off. 500k moves, seed 909, `-db-mode comp -p-comp 1.0 -p-any 0.0`, `-anc-samples 256`. All 16 convex arms and all 8 no-descent contiguous arms completed; the 8 descent×contiguous arms are ∼30× slower and are reported separately.

2.2 Convex half: main effects

factor (mean of 8)	gates removed	DB attempts	gates/hit	bits/splice
descent on / off	15,290 / 8,748	4,853,272 / 469,074	0.060 / 0.286	0.232 / 0.355
s_{db} 13 / 12	11,906 / 12,133	2,785,496 / 2,536,850	0.177 / 0.169	0.295 / 0.293
reshuffle on / off	12,284 / 11,754	2,641,481 / 2,680,865	0.172 / 0.173	0.288 / 0.299
curated on / off	12,077 / 11,961	2,629,091 / 2,693,255	0.159 / 0.187	**0.562 / 0.025**

The effects are additive with no sign reversals, and the best single cell is also the composite of all four: `s_db 12, convex, descent, reshuffle, curated` produced the smallest circuit of all sixteen (84,321 from 100,000) *and* the second-highest ancestry.

2.3 The mixing channel: descent is worth 600×

factor (mean of 8)	est_anc	cov	ent	reach	carriers	gen_r
descent on / off	**26,820 / 44**	0.456 / 0.037	0.668 / 0.085	0.448 / 0.025	0.807 / 0.093	0.859 / 0.331
s_{db} 13 / 12	13,371 / 13,492	0.245 / 0.248	0.373 / 0.381	0.236 / 0.237	0.445 / 0.455	0.591 / 0.599
reshuffle on / off	13,972 / 12,892	0.251 / 0.242	0.383 / 0.371	0.240 / 0.233	0.457 / 0.443	0.603 / 0.588
curated on / off	13,955 / 12,909	0.255 / 0.238	0.383 / 0.371	0.244 / 0.229	0.459 / 0.441	0.609 / 0.582

Here `est_anc` is the mean number of distinct *input* gates each current gate descends from (computed as desc_mean × m/size over K=256 tracers). So the descent arms have every gate carrying lineage from ∼28% of the 100k input, while the no-descent arms carry lineage from ∼44 gates — essentially nothing. **Without descent, COMP is a size-only pass.**

2.4 Descent is a transport lever, not a size lever

The two axes point in opposite directions per unit CPU:

	size	ancestry
descent	9.6 gates/s	17.9 anc/s
no descent	**58 gates/s**	0.29 anc/s
ratio	6× worse	60× better

The size trajectories explain it. Descent front-loads and exhausts; no-descent decays gently and *overtakes it on marginal rate* by move 500k:

moves	descent removed	Δ/50k	no-descent removed	Δ/50k
50k	5,159	5,159	1,771	1,771
100k	8,101	2,942	3,012	1,241
200k	11,325	1,257	5,314	974
300k	13,238	913	7,033	728
400k	14,687	614	8,319	578
500k	**15,679**	**427**	9,350	**533**

Consequence: for *compression at all costs* under a CPU budget, drop descent and spend the savings on moves. For compression *without compromising mixing*, descent is not optional.

2.5 Curated in COMP: the contract holds on size, fails on entropy

Curated is worth +1.0% on size — confirming the "COMP is regular-only" contract on that axis — but it also uses *2.4% fewer* attempts (it hits earlier in the cascade), gives +8.1% ancestry in all four matched descent pairs, and multiplies selection entropy by 22×. Its extra hits are size-neutral re-encodings, visible in gates/hit (0.159 with vs 0.187 without). It supplies 38–41% of successful splices under descent and 61–62% without. **Recommendation: reverse the contract** — the original was written on the size axis alone.

2.6 Contiguous: 8 matched no-descent pairs

Three points. First, convex wins size by 16× and transport by 7× in *every* pair. Second, contiguous costs 31× the wall time at essentially identical attempt counts (487k vs 470k), so the per-attempt canonicalization is ∼30× dearer — **the damage is window width, not the span cap**, since contiguous skips only 0.5–3.6% of attempts at the cap. Third, contiguous's one apparent win — per-splice entropy (0.766 vs 0.355) — **does not survive aggregation**: convex lands 2.4× more splices, so it delivers more *total* entropy in all 8 pairs (e.g. 23,021 vs 14,881 bits at s12/noshuf/cur).

2.7 Contiguous with descent

arm	final	removed	est_anc	carriers
s12 noshuf cur	98,959	1,041	195.0	0.228
s12 noshuf nocur	99,027	973	221.7	0.248
s12 shuf cur	97,372	2,628	4,269.9	0.585
s12 shuf nocur	97,325	2,675	**6,372.7**	0.606
s13 noshuf cur	98,997	1,003	190.2	0.221
s13 noshuf nocur	99,030	970	235.5	0.247
s13 shuf cur*	97,687	2,313	2,686.6	0.557
s13 shuf nocur*	97,748	2,252	3,650.3	0.576

cell	geom	final	rm	hit%	g/hit	bits	est_anc	carr	gen_r	wall
s12 noshuf nocur	ctg	99,586	414	2.56	0.033	0.004	3.5	0.009	0.142	3,852s
	cvx	91,520	8,480	5.86	0.306	0.035	37.0	0.079	0.318	152s
s12 noshuf cur	ctg	99,578	422	1.91	0.045	1.585	3.2	0.008	0.155	3,968s
	cvx	91,395	8,605	7.30	0.253	0.677	49.4	0.105	0.350	149s
s12 shuf nocur	ctg	99,273	727	3.65	0.041	0.003	10.0	0.024	0.212	4,993s
	cvx	90,790	9,210	6.63	0.296	0.034	42.4	0.093	0.312	113s
s12 shuf cur	ctg	99,291	709	2.74	0.053	1.467	8.9	0.021	0.228	5,033s
	cvx	90,650	9,350	8.03	0.252	0.671	69.1	0.133	0.378	160s
s13 noshuf nocur	ctg	99,580	420	2.35	0.037	0.004	2.6	0.007	0.100	4,634s
	cvx	91,877	8,123	5.35	0.320	0.034	25.3	0.057	0.287	172s
s13 noshuf cur	ctg	99,568	432	1.84	0.048	1.595	2.7	0.007	0.111	4,690s
	cvx	91,693	8,307	6.64	0.267	0.684	41.5	0.089	0.334	172s
s13 shuf nocur	ctg	99,414	586	3.31	0.037	0.003	9.2	0.022	0.218	5,540s
	cvx	91,132	8,868	5.87	0.320	0.036	34.1	0.079	0.308	141s
s13 shuf cur	ctg	99,295	705	2.46	0.059	1.465	8.0	0.019	0.218	5,728s
	cvx	90,956	9,044	7.10	0.273	0.668	52.7	0.111	0.360	185s
mean	**ctg**		**552**	2.60	0.044	0.766	**6.0**	0.015	0.173	**4,805s**
	cvx		**8,748**	6.60	0.286	0.355	**43.9**	0.093	0.331	**156s**

* stopped at 450k of 500k moves.

Descent lifts contiguous a lot in relative terms (removals 2.7–3.7×, ancestry up to 640×) but it remains far behind convex-with-descent (14,948–15,679 removed, est_anc 23,704–29,487). Two notes worth carrying: reshuffle is contiguous's *dominant* lever (est_anc 222 → 6,373, a 29× swing), and **curated *hurts* ancestry in contiguous** — negative in all four pairs — the exact opposite of its +8% in convex.

2.8 A bug: the shipped COMP sampler was inverted

`-p-convex-comp` shipped at **0.1** from the 2026-08-03 defaults commit, i.e. 90% contiguous, when the intent was the reverse. Every run taking the COMP default before 2026-08-04 spent ∼90% of its COMP DB budget on the geometry that loses on every axis above. Fixed to 0.9 in `46350179`. MIX's `-p-convex 0.4` was left alone: no MIX run had ever varied geometry, so there was no measurement behind it either way.

3 Part II: per-length structure

3.1 Method

With `-no-db-prefixes` the window length is drawn *uniformly* from $1..s_{db}$, so "hit rate at length k" is a genuine per-length conversion rate. With descent on, the same counter would be conditional on every longer length having already failed. Every table in this part uses the uniform draw, $s_{db} = 20$, 600k moves — about 30k independent attempts per length.

3.2 COMP, convex, four circuits: fat changes amplitude, not width

The compressible-fat hypothesis is confirmed on *volume*: MIX-grown material is 25–38% compressible, near-minimal material 1.0%, a 38× spread. But the *useful-width profile barely moves*: across that range the peak shifts by one gate and the median by two. This refuted our prior expectation that the fresh-material peak at 7 was an artifact of picked-over material. What fat buys is amplitude, plus a modest long tail (lengths 13–20 give 14.5% of `fat318`'s removals versus 0.5% of `base`'s).

len	base (100k fresh)	nR20 (179k minimal)	fat200 (203k)	fat318 (318k)
3	336	20	1,348	2,912
4	541	49	1,917	4,027
5	960	93	4,163	8,846
6	1,274	235	5,438	11,105
7	**1,686**	362	7,689	15,304
8	1,398	**433**	**8,769**	**17,522**
9	812	178	6,573	15,403
10	347	88	4,620	12,353
12	52	42	2,125	6,883
14	12	38	873	3,642
16	0	40	348	1,851
18	0	12	158	1,049
20	0	26	102	367
total	7,638	1,731	50,029	122,324
% of circuit	7.6%	**1.0%**	24.7%	**38.5%**
peak length	7	8	8	8
50% of mass by	7	8	8	9
share from len >14	0.0%	5.4%	3.3%	6.9%

Weighting by draw cost, the efficiency-optimal s_{db} is **8 for lean material and 10–12 for fat** — landing on the factorial's $s_{db} = 12$ winner and confirming it was not an artifact of one input.

3.3 COMP, contiguous, same four circuits

circuit	in	cvx out	cvx removed	ctg out	ctg removed	cvx/ctg
base	100,000	92,362	7,638 (7.6%)	99,612	388 (0.4%)	**19.7**×
nR20	179,132	177,401	1,731 (1.0%)	178,773	359 (0.2%)	4.8×
fat200	202,674	152,645	50,029 (24.7%)	186,706	15,968 (7.9%)	3.1×
fat318	317,813	195,489	122,324 (38.5%)	290,553	27,260 (8.6%)	4.5×

Convex wins everywhere, but the gap *narrows sharply with fat*: 19.7× on lean fresh material, 3–4.5× on MIX-grown circuits. Contiguous does find real compression when obvious fat exists; it collapses on picked-over material. Two structural differences from the convex ladder: contiguous *is* span-crushed above ∼11 at this s_{db} (skips exceed 50% by length 13–14 and reach 94–99% at 20, against convex's 7.45% maximum), and its useful width *does* move with the material (peak at 3, 9, 11, 10 across the four circuits, against convex's steady 7, 8, 8, 8). Contiguous wall times were 4,973–8,648s.

3.4 MIX: the DB move is a mean-reversion operator

Short windows almost always find a longer spelling and expand; long windows almost always find a shorter one and compress. Over the run, lengths 1–4 added 556k gates and lengths 5–20 clawed back 250k, net +313k. So **s_{db} in MIX is a net-growth-rate knob**, not a search-width knob: raising it dilutes expansion with self-compression. This is a second lever on growth alongside `p_mix`, which the layer-2 controller does not currently know about.

Each geometry has its own fixed point: convex+curated ≈ 5, convex−curated ≈ 5.5, contiguous+curated ≈ 6.5. That is why contiguous grew *more* (483,839 vs 412,797 final) despite adding about the same at lengths 1–2: it compresses less above the crossover.

len	hit%	net (− = growth)	len	hit%	net
1	99.78	−261,276	11	15.80	+21,964
2	99.02	−182,797	12	12.62	+21,237
3	97.53	−79,246	13	10.33	+20,719
4	91.15	−27,073	15	6.60	+17,047
5	75.21	**+175**	18	2.98	+10,279
6	67.09	+8,477	20	1.72	+6,949

Table 2: MIX, convex, curated, $s_{db} = 20$, 600k moves. The fixed point is at window length ≈ 5.

At $s_{db} = 20$ specifically, lengths 7–20 added **exactly zero** gates across all fourteen of them while removing 31,372; 98.6% of all expansion came from lengths 1–4 on 20% of the draws, and 77% from lengths 1–2 alone.

3.5 Hit rate by length: the full mode × geometry × curation cube

All eight cells, same circuit (`pre2_100k`), same budget, uniform draw over 1..20, ≈30k attempts per length.

	MIX				COMP			
len	cvx +cur	cvx −cur	ctg +cur	ctg −cur	cvx +cur	cvx −cur	ctg +cur	ctg −cur
1	99.78	99.75	99.79	99.70	0.00	0.00	0.00	0.00
2	99.02	98.83	98.23	97.95	0.00	0.00	1.60	**11.95**
3	97.53	96.79	91.45	83.91	13.87	6.71	12.33	9.83
4	91.15	85.47	79.22	76.71	19.59	13.64	5.66	5.37
5	75.21	70.10	64.23	61.73	16.64	16.64	2.62	2.71
6	67.09	62.43	54.75	49.87	17.15	17.32	1.01	1.03
7	45.50	39.42	42.89	39.72	8.39	8.78	0.52	0.50
8	35.49	23.57	32.89	31.47	2.94	2.87	0.34	0.25
9	25.52	15.06	24.36	24.53	1.17	1.16	0.12	0.10
10	20.14	9.72	15.82	14.24	0.36	0.39	0.04	0.05
11	15.80	6.58	11.77	8.60	0.16	0.18	0.01	0.01
12	12.62	3.85	7.56	2.42	0.04	0.03	0.00	0.00
13	10.33	2.44	6.33	1.70	0.02	0.01	0.00	0.00
14	8.10	1.55	4.64	0.75	0.01	0.00	0.00	0.00
15	6.60	0.95	3.81	0.37	0.00	0.00	0.00	0.00
16	4.76	0.60	2.77	0.13	0.00	0.00	0.00	0.00
17	3.87	0.45	2.27	0.10	0.00	0.00	0.00	0.00
18	2.98	0.21	1.45	0.07	0.00	0.00	0.00	0.00
19	2.15	0.18	1.20	0.02	0.00	0.00	0.00	0.00
20	**1.72**	**0.09**	0.84	0.01	0.00	0.00	0.00	0.00
overall	**36.27**	30.82	32.33	29.72	**4.02**	3.39	1.22	1.59

MIX converts ∼9× more often than COMP, which is just the acceptance rule (MIX takes any spelling, COMP requires strictly shorter; COMP's 0.00% at lengths 1–2 is structural). Half of all successful splices come from lengths ≤ 4 in every MIX arm (≤ 5 in COMP), on 20–25% of the draws; above length 7 the decay is close to geometric, ×0.75 per gate with curation and ×0.6 without.

Curation's value is entirely at the wide end, in both geometries. Nothing at length 1 (99.78 vs 99.75). In convex, 3.3× at length 12 and 19× at length 20; in contiguous the ratio is *larger* still — 3.1× at 12 and **84**× at 20 (0.84 vs 0.01). The curated store is a wide-window store.

But curation is net-negative for COMP contiguous — 1.22% against 1.59% overall, the only cell where it lowers the aggregate hit rate. It is driven entirely by length 2, where curation costs

cell	added	removed	net	max span-skip
MIX cvx +cur	562,415	249,618	+312,797	0.9%
MIX cvx −cur	324,931	94,309	+230,622	1.9%
MIX ctg +cur	493,073	109,234	+383,839	**70.4%**
MIX ctg −cur	263,530	21,097	+242,433	**94.2%**
COMP cvx +cur	0	7,638	−7,638	2.3%
COMP cvx −cur	0	7,454	−7,454	2.2%
COMP ctg +cur	0	388	−388	**99.3%**
COMP ctg −cur	0	390	−390	**99.4%**

11.95% → 1.60%. Removals are unchanged (388 vs 390), so this is substitution of equals rather than lost compression — but it is the same displacement effect that cost ancestry in the descent×contiguous arms of Part I, now visible on a third independent measure. Curation should be read as a convex-mode asset.

Finally, the span-skip column isolates what stops contiguous at high s_{db}: 70–94% of MIX contiguous attempts and *99%* of COMP contiguous attempts are rejected at the cap, against $\leq$ 2.3% for every convex cell.

4 Part III: MIX $s_{db}\times$ geometry at matched size

4.1 Design

Arms are compared at *matched final size*, not matched moves: MIX with pay-random grows, and net growth per unit work falls monotonically with s_{db}, so a fixed move budget would leave every arm at a different size and a low-s_{db} arm would "win" transport merely by being bigger. Each arm's move budget was calibrated from a measured growth pass so it lands at $\approx 2\times$ the 100k input.

4.2 Results

geo	s_{db}	moves	size	wall	added	removed	mv/kgate	bits/spl	tot bits	bits/kmv
cvx	3	**19,000**	201,432	**1s**	101,766	334	**187**	**3.789**	70,157	**3,692**
cvx	5	30,000	202,095	2s	103,501	1,406	294	3.100	80,364	2,679
cvx	7	44,000	204,755	3s	109,719	4,964	420	2.845	91,029	2,069
cvx	9	60,000	205,961	6s	116,686	10,725	566	2.752	100,330	1,672
cvx	12	86,000	205,515	12s	126,403	20,888	815	2.690	111,562	1,297
cvx	20	157,000	204,028	71s	138,910	34,882	1,509	2.632	**123,075**	784
ctg	3	21,000	199,925	1s	100,011	86	210	3.774	71,664	3,413
ctg	5	34,000	203,909	7s	104,259	350	327	3.055	75,846	2,231
ctg	7	48,000	205,271	42s	106,370	1,099	456	2.795	77,508	1,615
ctg	9	65,000	211,802	165s	114,305	2,503	581	2.676	83,563	1,286
ctg	12	93,000	217,039	559s	122,293	5,254	795	2.610	89,810	966
ctg	20	171,000	224,427	1,039s	135,165	10,738	1,374	2.550	99,570	582

4.3 The contiguous cost curve is non-monotone

s_{db}	3	5	7	9	12	20
ctg/cvx wall	1.1×	3.6×	12.6×	29.9×	**47.8**×	14.7×

The penalty peaks at $s_{db} = 12$ and then *falls*, because by $s_{db} = 20$ the span cap rejects ~70% of contiguous windows *before* canonicalization — the cap acts as a cheap early-out. The expensive regime is exactly s_{db} 9–12, which is where the shipped MIX default sat.

4.4 Verdict, and one thing this cannot answer

Convex at every s_{db} (cheaper and more total entropy); low s_{db} for expansion efficiency (187 vs 1,509 moves per 1,000 gates grown).

But **est_anc stays 1–10 in all twelve arms** — pure MIX expansion to 2× transports essentially no ancestry at any setting. The braiding seen in long p_mix runs (est_anc 12k–44k, carriers 0.98) comes from sustained churn over millions of moves, not from the expansion leg. So this sweep answers "which s_{db} expands best" and *not* "which braids best at held size"; the latter needs long runs with size pinned.

Under the phase-B reframe, where spread is judged in *absolute* gates, reach of 0.001–0.004 on a 205k circuit is 200–820 positions, which is the same order as the ~850 that reframe calls meaningful. So these arms are not as inert as the fractional numbers suggest.

5 Part IV: re-reading the p_mix dose-response

Eight runs at 2M moves, single DB configuration, only `p_mix` varying:

p_mix	size	est_anc	cov	reach	carriers	card	anc ÷ (mv/gate)
0.05	128,355	44,302	0.519	0.510	0.984	115.2	2,840
0.10	150,699	38,844	0.461	0.446	0.988	100.6	2,921
0.15	176,429	32,181	0.381	0.369	0.991	83.1	2,848
0.20	202,674	26,141	0.313	0.300	0.993	67.4	2,649
0.25	235,577	22,119	0.268	0.254	0.994	57.0	2,605
0.30	277,404	17,719	0.218	0.204	0.995	45.6	2,458
0.35	317,813	14,896	0.187	0.172	0.996	38.3	2,368
0.40	369,987	12,220	0.158	0.141	0.997	31.4	2,259

Read raw, this looks like a 3.6× collapse in transport as p_mix rises — a strong argument for keeping it low. **It mostly isn't.** All eight got 2M moves at different final sizes, so moves-per-gate falls 15.6 → 5.4, a 2.9× dilution that nearly accounts for the whole effect. Normalised (last column), the decline is **26% across the entire range**. The one unambiguous result: carriers saturates at ≥ 0.984 everywhere, so all the differentiation is in depth, none in breadth.

6 Part V: the GSS profile

6.1 Specification

The DB settings for running fmix on a gadgetized sliced sandwich. Exposed as `-gss`; explicit flags win; composes with `-phase-a`.

	COMP-DB	MIX-DB
curated	on	on
descent	**on**	**off**
`p_mingen`	0	0.5
geometry	convex 95% / contiguous 5%	convex 50% / contiguous 50%
s_{db} convex	12	6
s_{db} contiguous	6	6

`-gss` deliberately does *not* set `p_mix`: the MIX/COMP balance is the layer-2 controller's lever, and the profile is intended to be the right per-mode setting at every p_mix.

6.2 Evidence for each setting

COMP convex $s_{db} = 12$ with descent is exactly the factorial's best cell, and $s_{db} = 12$ matches the measured productive band on GSS-like material (`nR20_mixed`: peak at 8, lengths >14 worth 5.4%). Capping contiguous at $s_{db} = 6$ keeps it inside both the affordable regime (the ctg/cvx cost ratio is 3.6× at 5 but 47.8× at 12) and the span-safe one (skips begin around length 10); at 5% weight it buys geometric diversity cheaply, and on all-g57 material it still converts at 52–96% for lengths 3–9. MIX at $s_{db} = 6$ without descent converts on ≈88% of draws (99/99/97/91/75/67% across lengths 1–6) at the cheapest width.

6.3 The profile is g57-preserving, by design

arm	comp=	g57=	shaped=	polf=
M20_cvx_cur (MIX)	398,904	398,904	398,904	**0.000**
MS_cvx_s12 (MIX)	175,555	175,555	175,555	**0.000**
LB_base (COMP)	61,691	61,691	61,691	**0.000**
LB_nR20 (COMP)	176,118	176,118	176,118	**0.000**
f32_12_desc_cvx_shuf_cur (COMP)	69,340	69,340	69,340	**0.000**

`comp` = `g57` = `shaped` holds exactly in every DB-only run, and `polf` — the structure-breakdown meter — is identically zero. The store emits g57-form words, so every DB splice re-spells a g57 word as another g57 word. This is intentional: the profile's job is re-spelling, and breaking g57 form is a separate concern requiring the twist family, not the store.

6.4 Implementation

Two mechanisms had to be added.

Geometry is now drawn once per round, before the length. It used to be drawn inside `sample_window` *after* the length was fixed, which made a geometry-conditional length inexpressible and let the best-of-`litter_samples` selection compare windows drawn under different geometries. New flags `-s-db-ctg` and `-s-db-comp-ctg`; resolution is most-specific-first (mode+geometry → mode → base), with 0 falling through rather than clamping.

Descent is now per-mode, via `-db-prefixes-mix` and `-db-prefixes-comp` (unset inherits the global `-db-prefixes`), because the `-p-mix` overlay runs both modes in one process and they want opposite settings.

Cleanup in the same change: `DbSample::Mixed` and `DbSample::parse` were removed (nothing constructed or called them since the sampler knobs were split per mode), and the DB banner now prints the *effective* per-mode settings rather than the base knobs.

7 Part VI: measurement traps

We think this is the most reusable section. Every one of these cost real runs.

Shipped defaults move under a rebuild. `-db-prefixes` and `-curated` both flipped ON between two builds of the same binary. A launch script copied forward therefore silently changed the measurement. This is fatal for per-length work: uniform draw gives a per-length conversion rate, descent gives a quantity conditional on longer lengths failing. *Always pin every knob a measurement depends on.* Sanity check without reading the banner: under a uniform draw a 600k-move run gives $\approx$ 600k/s_{db} attempts per length; descent gives $\approx$600k at s_{db} alone.

The COMP overrides shadow the base knobs. `s_db_comp` (12) and `p_convex_comp` (0.9) ship as concrete values, and `active_s_db`/`active_p_convex` prefer them whenever the live mode is COMP — despite doc comments promising sentinel fall-through. A run passing `-db-mode comp -s-db 20` silently gets 12. Detected because the COMP per-length tables stopped at length 12 while the MIX ones reached 20.

`-target-size` does not cap DB-driven growth. With `-p-db 1.0` the DB move fires every round and the size brake does not gate it. A MIX sweep intended to stop at 200k ran $s_{db} = 3$ to 3,726,036 gates and OOM-killed six sibling jobs. Match sizes by *budgeting moves* from a measured growth calibration instead.

Hit rate is not a compression proxy. `nR20_mixed` has the highest hit rates anywhere (47–77% at lengths 3–6) and almost no removals; `pre2_100k` has the lowest (14–19%) and removes far more. Hit rate tracks *g57-form fraction* (what the store can match); removal tracks *compressible fat*. Reading one for the other ranks these two circuits exactly backwards.

Per-splice entropy inverts under aggregation. Contiguous wins bits/splice by 2.2× and loses total entropy in all 8 matched pairs, because it lands 2.4× fewer splices. Always report both.

Budgeting by moves flatters whichever arm is slower per move. Descent "wins" size by 75% at matched moves and loses by 6× at matched CPU. State the budget axis explicitly.

8 Open questions

1. **Matched-work no-descent** ($\approx$5M moves, $\approx$25 min). The trajectories predict no-descent overtakes descent on size at equal CPU; whether its ancestry percolates is unknown and would settle the COMP configuration.

2. s_{db} **9/10 with descent in COMP.** The factorial is monotone decreasing (12 beats 13) and 92% of convex removals come from lengths ≤ 9, so the optimum may be lower than 12.

3. **MIX braiding at held size.** Part III measures expansion, not transport. Long runs with size pinned would say which s_{db} braids best.

4. **Layer 2 and s_{db}.** MIX s_{db} is a second growth lever the controller does not model.

5. **Why does curation cost COMP contiguous its length-2 hits?** 11.95% $\rightarrow$ 1.60% is too large to be noise and has no analogue in convex. The cascade prefers a curated match when one exists; at length 2 in contiguous that appears to displace a regular match that would have been taken. Removals are unaffected, so nothing is lost on size — but the mechanism is not understood, and it is the third measure on which curated hurts contiguous.